MIL-C-20313
21 November 1951

MILITARY SPECIFICATION

CLOTH, COTTON, CARTRIDGE

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

Army 50-11-65A
19 August 1941

This specification consists of this cover sheet, Specification 50-11-65A, dated 19 August 1941, and Amendment 2, dated 17 October 1944, attached hereto, without modification.

When a request for this specification is received by a supplying activity it will be necessary to attach this cover sheet to the pertinent specification before issue

Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer .

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U. S. ARMY
SPECIFICATION

No. 50-11-65A

AUGUST 19, 1941

SUPERSEDING
No. 50-11-65
June 19, 1933

CLOTH, COTTON, CARTRIDGE

A. APPLICABLE SPECIFICATIONS.

A–1. The following specifications, of the issues in effect on the date of invitation for bids, form a part of this specification:

Federal Specification CCC–T–191, Textiles; General Specifications, Test Methods.

U. S. Army Specification No. 100–2, Standard Specifications for Marking Shipments.

B. GRADES.

B–1. This specification covers four grades, A, B, C, and D.

C. MATERIAL AND WORKMANSHIP.

C–1. *Material.*—

C–1a. Cotton cartridge cloth shall be made of thoroughly cleaned cotton, free from waste.

C–1b. Grade C cloth shall be manufactured from material commercially known as combed lawn cloth, 40 inch, 88 by 80 threads, 6.90 yards per pound.

C–2. *Workmanship.*—

C–2a. Cotton cartridge cloth shall be manufactured in such a manner as to be free from avoidable imperfections and defects or blemishes that might affect its serviceability. Pieces shall be not less than 20 yards in length.

C–2b. Cotton cartridge cloth shall be bleached.

C–2c. Grade C cloth shall be very slightly sized with starch.

C–2d. Grade C cloth shall be manufactured in such a manner that the finished width shall not exceed the original greige width.

C–2e. Grade D cloth shall be pre-shrunk and mercerized. It shall be a four harness twill weave to the right, three up and one down.

D. GENERAL REQUIREMENTS.

D–1. *Color.*—White.

D–2. *Ash.*—Maximum, 0.2 percent.

D–3. *Acidity or alkalinity.*—

D–3a. *Reaction to litmus paper.*—Neutral.

D–3b. *Organic acidity or alkalinity.*—Maximum, 0.1 percent.

D–4. *Width.*—

Minimum, inches	36
Maximum, inches	[1] 42

[1] Unless otherwise specified.

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E. DETAIL REQUIREMENTS.

E–1. Cartridge cloth shall conform to the requirements shown below for the grade specified:

	Grade A	Grade B	Grade C	Grade D
Weight per square yard: Maximum, ounces	5.0	3.0	2.1	8.2
Thread count:				
Warp, minimum			87	116
Warp, maximum			93	
Fill, minimum			76	56
Fill, maximum			82	
Breaking strength:				
Warp, minimum, pounds	50	30	28	180
Fill, minimum, pounds	50	30	25	130
Stretch:				
Warp, maximum, percent				12.5
Fill, maximum, percent				12.5
Water extract: Maximum, percent	0.5	0.5	2.0	0.5

F. METHODS OF SAMPLING, INSPECTION, AND TESTS.

F–1. *Inspection.*—Inspection shall be at the manufacturer's plant unless otherwise specified in the contract.

F–2. *Size of lots.*—Maximum, 100,000 square yards.

F–3. *Sampling* shall be done as follows: Select 10 percent, but not more than 10 in number, of the bales or boxes comprising the lot. Remove one bolt from each selected bale or box, unroll at least 50 feet from the bolt, and cut a sample of two running yards from the bolt. Label each sample so that the bale or box from which it was taken can be identified. Make a composite sample of the lot, consisting of approximately 1 yard from each of the primary samples. Label all samples, giving the name of the material, grade, manufacturer, plant, purchase order, and the number of yards in the lot. All acceptance tests shall be made on the composite sample representative of the lot. Hold the primary samples for possible future examination should the composite sample fail to meet the requirements.

F–4. *Laboratory* examination shall be made as follows:

F–4a. *Ash.*—Place a weighed sample of approximately 2 grams, consisting of equal portions from each piece of cloth in the composite sample, in a tared platinum or porcelain crucible, and add enough concentrated nitric acid to moisten the entire sample. Digest on a steam bath for 1 hour, then heat carefully over a flame so as to avoid any loss. When the contents of the crucible are charred, heat to dull redness until all carbonaceous matter has been burned off. Cool in a dessicator and weigh. Calculate the percentage of ash from the weight of the residue.

F–4b. *Acidity or alkalinity.*—

F–4b (1). *Reaction to litmus.*—Cut 10 pieces of cloth, weighing approximately 0.5 gram each from various portions of the sample. Weigh these pieces together, transfer them to a 250 cc. beaker, and treat with 150 cc. of boiling water for 30 minutes. Keep the beaker

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covered with a watch glass during the period of boiling. In case the volume of the water becomes less than one-half of the original volume, add sufficient neutral distilled water to restore it to the original volume. At the end of 30 minutes again bring the volume of the water to 150 cc. by the addition of neutral distilled water, cool, and add pieces of red and blue litmus paper which have been found to be sensitive to N/1000 sodium hydroxide and N/1000 sulphuric acid, respectively, when immersed in these for three minutes. At the end of three minutes, note if the extract of the sample is neutral as indicated by the failure of either piece of litmus paper to be changed to the color produced by immersion of the paper in N/1000 alkali or acid.

F–4b (2). *Organic acidity or alkalinity.*—If the litmus paper is unchanged at the end of three minutes, indicating the absence of inorganic acidity or alkalinity, test for organic acidity or alkalinity as follows: Filter the solution and titrate with approximately N/10 alkali or acid, using phenolphthalein as the indicator. Calculate any acidity or alkalinity to percentage of acetic acid or sodium hydroxide.

F–4c. *Width.*—Determine as directed by Specification CCC–T–191.

F–4d. *Weight.*—Determine as directed by Specification CCC–T–191.

F–4e. *Thread count.*—Determine as directed by Specification CCC–T–191.

F–4f. *Breaking strength.*—Determine as directed by Specification CCC–T–191 for the grab method.

F–4g. *Stretch.*—Prepare test specimens similar to those used for determining breaking strength. Test each by inserting it in the jaws of the testing machine with a distance closely approximating one inch between the jaws. Carefully pull the jaws apart, and when the hand on the dial just starts to indicate tension measure the distance between the jaws by means of inside calipers and a micrometer or a steel rule and reading glass. Apply a load of 50 pounds and again measure the distance between the jaws. Calculate the increase in distance to percentage of the initial distance and calculate from these values the average percentage stretch in each direction.

F–4h. *Water extract.*—Cut equal portions from each piece of cloth in the composite sample so as to obtain a sample weighing approximately 5 grams. Weigh the pieces together to the nearest 0.01 gram and immerse them for 1 hour in 150 cc. of boiling distilled water. If the volume of water becomes less than one-half the original volume, add sufficient distilled water to replace that lost by boiling. Filter the solution, again treat the sample with 150 cc. of boiling distilled water for 1 hour, and filter. Transfer the combined filtrates to a tared beaker, and evaporate to dryness. Heat the beaker and residue at 100° C. for 1 hour, cool in a desiccator and weigh. Calculate the weight of residue to percentage of water extract in the sample.

G. PACKING AND MARKING.

G–1. *Packing.*—Cotton cartridge cloth shall be furnished in bolts tied with tape and packed in the form of bales or in boxes. Bolts packed in bales shall be wrapped in waterproofed paper and then in burlap. Bolts, taped but unwrapped, may be packed in wooden

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or plyboard boxes lined with waterproofed paper. Boxes shall be so constructed as to insure acceptance by common or other carriers, for safe transportation, at the lowest rate, to the point of delivery. A bale or box shall contain not more than 200 pounds of cloth and shall be of sufficiently rigid construction to withstand stacking 10 feet high in storage.

G–2. *Marking.*—Each bale or box shall be marked in accordance with U. S. Army Specification No. 100–2, with additional markings giving the net weight and lot number of the contents.

H. NOTES.

H–1. *Uses.*—The cotton cartridge cloth covered by this specification is intended for use in the manufacture of bags for loading propellent charges.

H–2. The use of this specification, whenever applicable, is mandatory on all procuring agencies of the Army.

H–3. If the composite sample representative of the lot fails to pass the inspection tests, the manufacturer shall have the option of having analysis of each primary sample made at his own expense. The manufacturer may then replace defective portions of the lot represented by the primary samples which fail to meet the requirements, and again submit the lot for inspection and acceptance, provided that the markings on the container are such that complete replacement of defective portions of the lot can be made to the satisfaction of the inspector.

NOTICE.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may be in any way related thereto.

NOTE.—Copies of this specification may be obtained from the Office, Chief of Ordnance, Washington, D. C.

[No. 50–11–65A]

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